

14. Suppose that there are $n = 2^k$ teams in an elimination tournament, where there are $n/2$ games in the first round, with the $n/2 = 2^{k-1}$ winners playing in the second round, and so on. Develop a recurrence relation for the number of rounds in the tournament.

- *28. Suppose someone picks a number x from a set of n numbers. A second person tries to guess the number by successively selecting subsets of the n numbers and asking the first person whether x is in each set. The first person answers either “yes” or “no.” When the first

person answers each query truthfully, we can find x using $\log n$ queries by successively splitting the sets used in each query in half. Ulam’s problem, proposed by Stanislaw Ulam in 1976, asks for the number of queries required to find x , supposing that the first person is allowed to lie exactly once.

- a) Show that by asking each question twice, given a number x and a set with n elements, and asking one more question when we find the lie, Ulam’s problem can be solved using $2 \log n + 1$ queries.
- b) Show that by dividing the initial set of n elements into four parts, each with $n/4$ elements, $1/4$ of the elements can be eliminated using two queries. [*Hint: Use two queries, where each of the queries asks whether the element is in the union of two of the subsets with $n/4$ elements and where one of the subsets of $n/4$ elements is used in both queries.*]
- c) Show from part (b) that if $f(n)$ equals the number of queries used to solve Ulam’s problem using the method from part (b) and n is divisible by 4, then $f(n) = f(3n/4) + 2$.
- d) Solve the recurrence relation in part (c) for $f(n)$.
- e) Is the naive way to solve Ulam’s problem by asking each question twice or the divide-and-conquer method based on part (b) more efficient? The most efficient way to solve Ulam’s problem has been determined by A. Pelc [Pe87].